

Interference of Light:

An important characteristic of light waves is their ability, under certain circumstances, to interfere with one another. Most people observe some type of optical interference every day, but do not realize what is occurring to produce this phenomenon. One of the best examples of interference is demonstrated by the light reflected from a film of oil floating on water. Another example is the soap bubble illustrated in Figure 1 that reflects a variety of beautiful colors when illuminated by natural or artificial light sources.

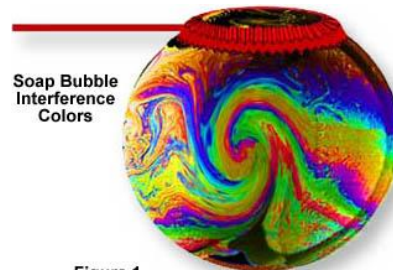
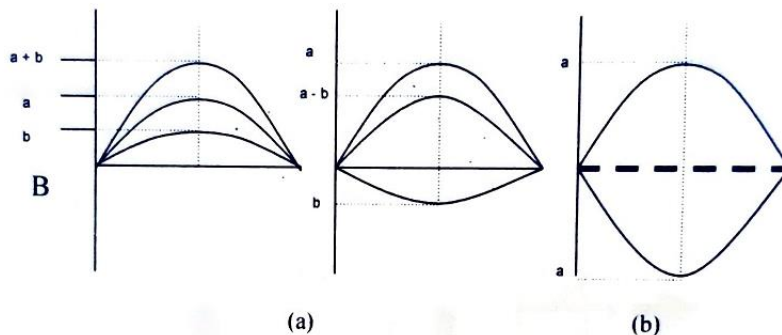


Figure 1

In physics, when two or more waves combine at a particular point, they are said to interfere, and this phenomenon is known as Interference.

Hereby interference refers to the physical effects of two or more wave trains to superposition. Here the resultant wavefront is strongly dependent on the relative phases of the interfering waves. There are two types of interference –

- a) Constructive Interference
- b) Destructive interference



(a) Wave shape for constructive interference. (b) Wave shape for destructive interference.

a) Constructive interference

It occurs at any location along the medium where the two interfering waves have a displacement in the same direction. For example, if at a given instant in time and location along the medium, the crest of one wave meets the crest of a second wave, they will interfere in such a manner as to produce a "super-crest." Similarly, the interference of a trough and a trough interfere constructively to produce a "super-trough."

b) Destructive interference

It occurs at any location along the medium where the two interfering waves have a displacement in the opposite direction. For example, the interference of a crest with a trough is an example of destructive interference. Destructive interference has the tendency to decrease the resulting amount of displacement of the medium.

Conditions for Interference:

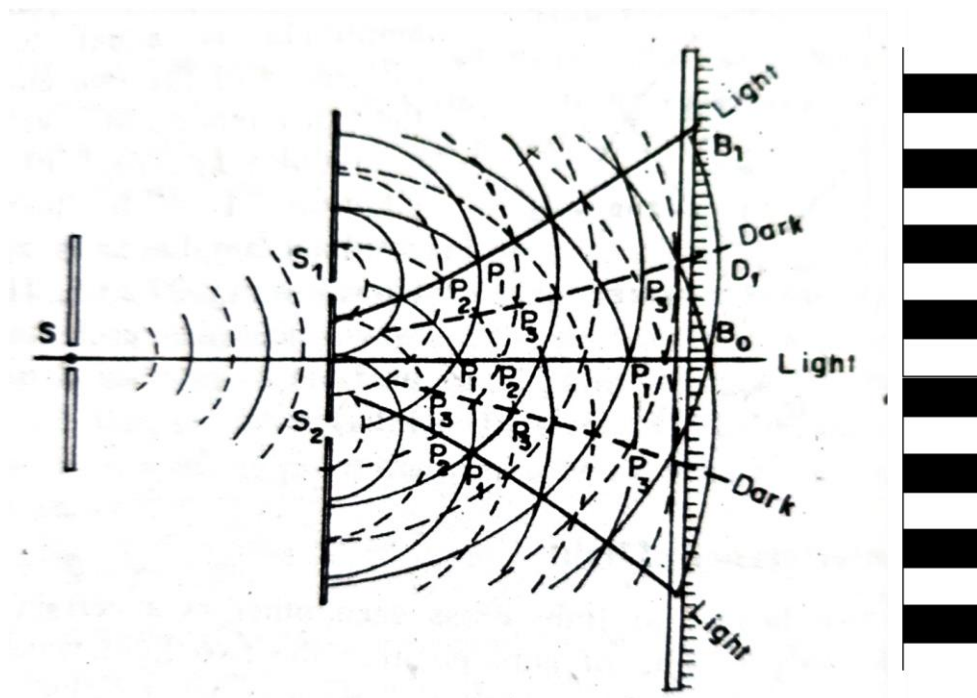
1. The light sources should be coherent.
2. The of two waves producing interference should be equal or almost equal.
3. The sources should be very close to each other.
4. The sources should be very narrow.

Young's Double-Slit Experiment:

Historically, the phenomenon of the interference of light was first demonstrated by Thomas in about 1801 by a simple experiment. Young allowed sunlight to pass through a pin hole S and then at some distance through two sufficiently close pin holes S_1 and S_2 in an opaque screen. Finally, the light was received on a screen on which he observed an uneven distribution of light intensity consisting of many alternate bright and dark spots.

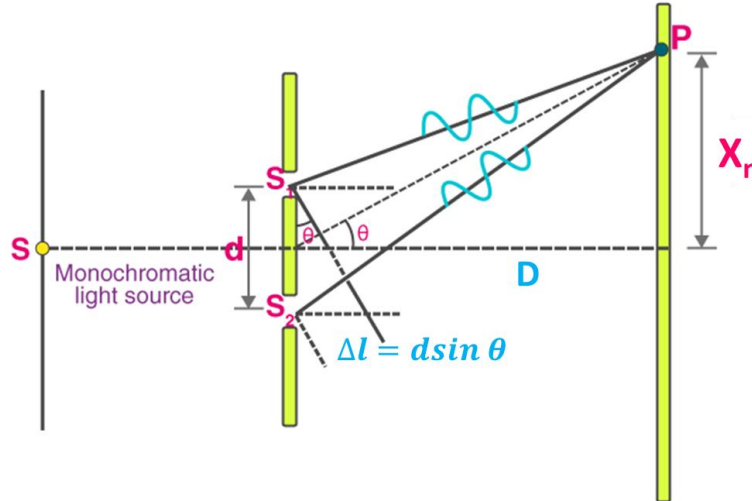
The original Young's double-slit experiment used diffracted light from a single source passed into two more slits to be used as coherent sources. Lasers are commonly used as coherent sources in the modern-day experiments.

According to this experiment, the slit S sends spherical wave front. Since the slits S_1 and S_2 are equidistant from S so the same wave front will reach S_1 and S_2 . The points S_1 and S_2 on this wave front emit secondary waves those are in phase with each other. So, the secondary waves emitted from the slits S_1 and S_2 are coherent as their frequency and amplitudes are same. Now the superposition of two waves emitted from S_1 and S_2 produces interference and shows the bright and dark fringes at the screen as like the figure.



Theory of Fringes:

According to the figure below, D is the distance between the slits and the screen, d is the separation between the slits of S_1 and S_2 , where $D \gg d$. the angle θ serves as the locator for the point P where two rays combined and produced n^{th} bright fringe (or dark fringe). Suppose X_n is the distance of the n^{th} bright fringe from the central bright fringe on the screen.



So, we can write the path difference between two rays from S_1 and S_2 –

$$\Delta l = d \sin \theta$$

Assume θ is small enough for what we can approximate:

$$\sin \theta \approx \theta \approx \tan \theta = \frac{X_n}{D}$$

$$\text{So, } \Delta l = d \tan \theta$$

$$\Delta l = d \frac{X_n}{D}$$

When two waves combine with each other in same phase, then the constructive interference occurs and as a result bright fringe formed. In this case, the phase difference δ between two waves will be $2\pi n$ and so that, the path difference –

$$\Delta l = n\lambda \quad \left[\text{Phase Difference, } \delta = \frac{2\pi}{\lambda} \times \Delta l \right]$$

$$\text{Where, } n = 0, 1, 2, 3, \dots \dots \dots$$

Again, when destructive interference occurs, then fringes are found and generally it happens when two waves combine in opposite phases i.e., when the phase difference between waves will be $(2n + 1)\pi$ and also the path difference –

$$\Delta l = (2n + 1) \frac{\lambda}{2} \quad \left[\text{Phase Difference, } \delta = \frac{2\pi}{\lambda} \times \Delta l \right]$$

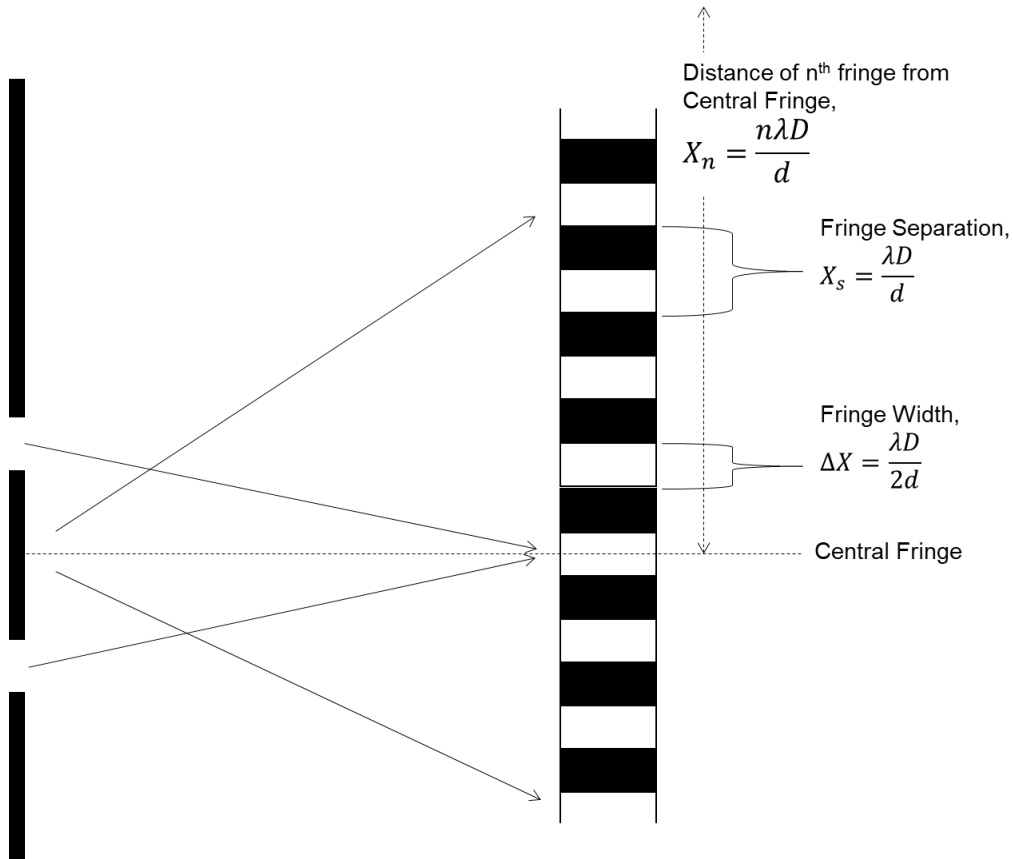
$$\text{Where, } n = 0, 1, 2, 3, \dots \dots \dots$$

Applying bright fringe condition, we get –

$$d \frac{x_n}{D} = n\lambda$$

The distance between n^{th} bright fringe from the central bright fringe is –

$$X_n = \frac{n\lambda D}{d}$$



Therefore, the fringe separation of two consecutive bright fringes will be as –

$$X_s = X_{n+1} - X_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d}$$

$$X_s = \frac{\lambda D}{d}$$

And the fringe width will be –

$$\Delta X = \frac{\lambda D}{2d}$$

Applications of Young's Double Slit Experiment:

1. Determination of wavelength of monochromatic light by using the equation, $X_n = \frac{n\lambda D}{d}$
2. Determination of refractive index of medium by using following equations –

$$\mu = \frac{X_s d}{l D} + 1$$

Where, l = thickness of the medium

Problem-1: In a Young's Double-Slit experiment, the separation between the sources are 0.18mm . the fringes are observed on a screen 90cm away. If with certain monochromatic source of light, the third bright fringe is situated at a distance of 8.1mm from the central bright fringe, find the wave length of light.
[Answer: $5400\text{\AA}$.U.]

Problem-2: The phase difference between two points on a wave is π , what is the path difference between them?
[Answer: $\lambda/2$]

Problem-3: In Young's Double-Slit experiment the distance between two slits is 2.0mm and the distance between slits and screen is 1m . The wavelength of the light used is $5900\text{\AA}$.U. Find the separation between two consecutive fringes.
[Answer: 0.295mm]

Problem-4: Interference fringes are produced on a screen kept at 1m away from two slits 0.4mm apart. The wavelength of light used is $5000\text{\AA}$. Find the distance between the centers of two consecutive bright and dark fringe.
[Answer: 0.625mm]

Problem-5: Light from a sodium vapour lamp ($\lambda = 589\text{nm}$) forms an interference pattern on a screen 0.8m from a pair of slits. The bright fringes in the pattern are 0.35cm apart. What is the slit separation?
[Answer: 0.135mm]

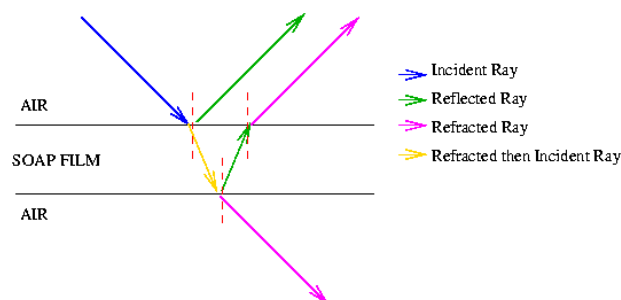
Problem-6: If 2cm is the separation between slits and 10cm is the distance between the screen and the coherent sources of a Young's experiment. In this arrangement is a glass plate of thickness 0.2cm is palced and the fringe separation is found to be 0.5cm . Calculate the refractive index of that glass plate.
[Answer: 1.5]

Problem-7: A thin sheet of Mica ($\mu = 1.58$) is used to cover one slit of the double-slit arrangement. The central point on the screen is now occupied by what to be the seventh bright fringe. If the wavelength of the light used is $5500\text{\AA}$.U. What is the thickness of the Mica sheet? [Answer: $6.67 \times 10^{-4}\text{cm}$]

$$\text{Hints: } l(\mu - 1) = n\lambda$$

Interference from Thin Films:

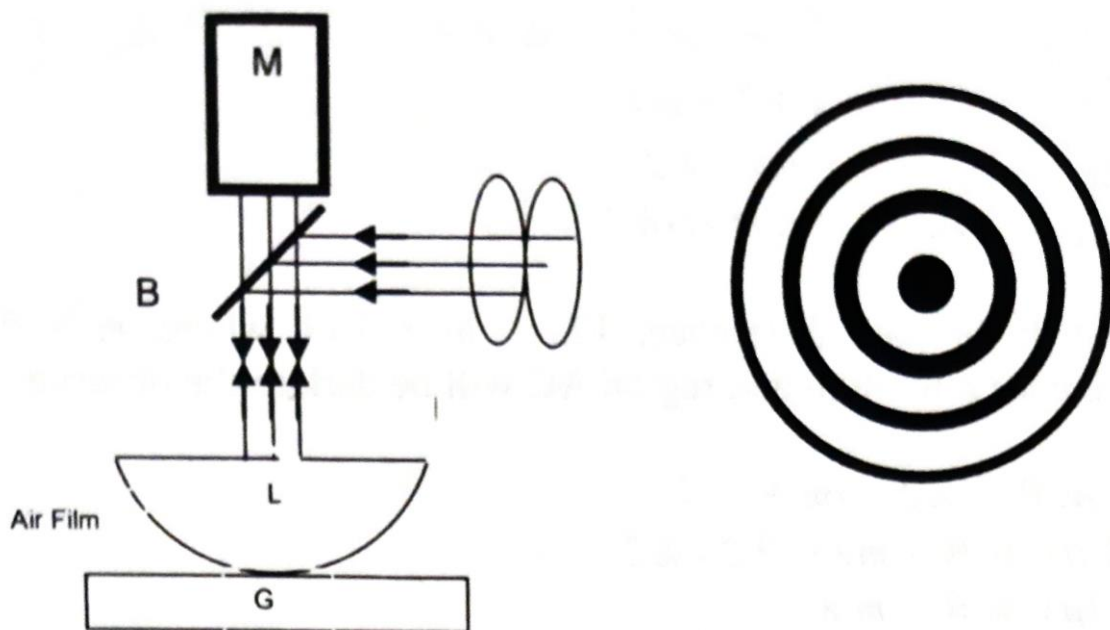
Thin-film interference is a natural phenomenon in which light waves reflected by the upper and lower boundaries of a thin film interfere with one another, either enhancing or reducing the reflected light. Thin-film interference explains the multiple colors seen in light reflected from soap bubbles and oil films on water. It is also the mechanism behind the action of antireflection coatings used on glasses and camera lenses.



Production of Newton's Ring:

When a plano-convex lens of long focal length is placed on a plane glass plate, a thin film of air is enclosed between the lower surface of the lens and upper surface of the glass plate. The thickness of the air film is very small at the point of contact and gradually increases from the center outwards. The fringes produced with monochromatic light are circular. The fringes are concentric circles, uniform in thickness and with the point of contact as the center. When viewed with the white light, the fringes are colored. With the monochromatic light, bright and dark circular fringes are produced in the air film.

In figure, S is a source of monochromatic light at the focus of the lens L. A horizontal beam of light falls on the glass plate B at an angle of 45° . The glass plate B reflects normally a part of the incident light towards the air film enclosed by the lens L and plane glass Plate, G.



Light rays reflected upward from the upper surface of the glass plate, G and lower surface of the lens, L that means that the rays reflected upward from the top and bottom surface of the air film that enclosed by the plane glass plate, G and lens L. The reflected rays then superimpose each other with a path difference depending upon the air film's thickness in between. Due to interference of these rays, dark and bright circular fringes are seen with the monochromatic light, which are actually known as Newton's rings. These rings are circular because the air film is symmetrical about the point of contact, O of the lens. These rings can be observed by means of a low power traveling microscope, M and that their diameter can also be measured in this manner.